

KAVERI 2.0 – Scalify Object Storage: Sizing Basis & Justification

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Background

Stamps and Registration department is having KAVERI 2.0 a registration system. KAVERI 2.0 is not merely a citizen-facing e-Governance application — it is a primary revenue collection system for the Government of Karnataka, processing stamp duty and registration fee payments across Document Registration, Firm Registration, and Marriage Registration. Across the last four fiscal years, the platform has processed revenue exceeding ₹1.14 lakh crore, underscoring that any downtime, capacity shortfall, or data-integrity risk on KAVERI is a direct risk to State revenue collection — not simply an IT service disruption.

₹1,14,099 Cr Total revenue, All categories — 4 years (2023-24 to 2026-27)	₹69,574 Average revenue per Document Registration transaction	+162% Document Registration revenue growth, 2026-27 vs 2025-26	₹156 Cr / day Est. daily revenue run- rate, Document Registration (2026-27)
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Document Registration is overwhelmingly the dominant revenue stream, accounting for effectively the entire ₹1.14 lakh crore total, at an average of ₹69,574 realized per registration transaction. Registration volume and revenue have both accelerated sharply in 2026-27, with revenue up 162% over the prior year and daily throughput of approximately 22,464 registrations — translating to a current run-rate of roughly ₹156 crore in Document Registration revenue processed per day.

At this scale, even a single day of unplanned downtime represents a material risk to State revenue realization, quite apart from the citizen-service and reputational impact. This is the core justification for treating KAVERI 2.0's infrastructure resilience, storage capacity, and cloud/DR investment — as detailed in the accompanying technical documents — as a revenue-protection priority, not a discretionary IT upgrade for a specific department.

1. Purpose

This document sets out the technical basis and justification for the Scality object-storage capacity requested for the KAVERI 2.0 application. It supplements the formal provisioning request submitted to the DPAR (e-Governance), State Data Centre (SDC) and provides the underlying methodology, module-wise derivation, and assumptions used to arrive at the storage figures, so that the request can be independently reviewed and validated. KAVERI 2.0 is a major state revenue system.

Document Registration			Firm Registration		Marriage Registration	
Registration Count	No of EC and CC	Total Revenue in Crores	Registration Count	Total Revenue in Rupees	Registration Count	Total Revenue in Rupees
25,74,690	20,78,990	1,60,78,00,00,000	0	0	2,004	1,16,999
27,17,602	40,09,285	1,92,00,00,00,000	9	875	1,78,103	1,86,11,726
29,07,144	47,37,530	2,17,69,00,00,000	6,465	6,93,263	2,15,424	2,26,74,347
81,99,436	1,08,25,805	5,70,47,00,00,000	3,138	3,23,508	1,05,433	1,02,55,966
		11,40,94,00,00,000		10,17,646		5,16,59,038

2. Scope

The scope of this sizing exercise covers primary object storage for documents and images generated by KAVERI 2.0 across:

- Part I — Modules currently live in production.
- Part II — e-Stamp, pending separation into an independent module with its own transaction stream.
- Part III — Paperless Registration, pending statewide roll-out across all 30 districts.

This document does not size the underlying database, application, or backup/DR storage tiers; it is limited to the Scality object-storage tier used for document and image repositories.

3. Sizing Methodology

3.1 Derivation Approach

For each module, the daily storage requirement is derived from the historical transactions done in the system till date as per the data provided by the application/functional team. This data was used to derive an average object size per transaction, using the average daily transaction volume for that module:

$$\text{Avg. Object Size per Transaction} = (\text{Size Requirement per Day}) \div (\text{Avg. Number of Transactions per Day})$$

This derived figure is used in Section 5 as a sanity-check basis against the nature of each module (e.g. a scanned paper registration document is expected to be materially larger than an EC or CC extract), rather than providing the daily GB figures as unexplained totals.

3.2 Projection Approach

Monthly, 6-month, and 12-month figures are derived by straight-line (linear) projection of the daily figure, using a 30-day month:

- $\text{Size / Month} = \text{Size / Day} \times 30$
- $\text{Size / 6 Months} = \text{Size / Month} \times 6$
- $\text{Size / 12 Months} = \text{Size / Month} \times 12$

No compounding transaction-growth factor has been applied in this projection; the figures represent storage consumption at the current observed transaction volumes held constant over the period. Where actual volumes grow, consumption will exceed these figures — see Section 8 (Risks & Considerations).

4. Assumptions

- A 30-day month is used uniformly for monthly projections; calendar-month variation is not separately modelled.
- 1 TB is treated as 1,000 GB (decimal convention), consistent with the source sizing sheet.
- Figures reflect average daily transaction volumes as furnished by the functional/application team, not observed peak-day volumes.
- Each module's storage is treated independently; no cross-module deduplication is assumed.
- Figures cover primary object storage only — replication, erasure-coding overhead, backup copies, and archival/DR copies are excluded from the base figures and addressed separately in Section 7.

5. Detailed Sizing by Module

5.1 Part I — Live Modules

Modules currently live and generating storage load in production. Individual 6-month/12-month projections below are computed by straight-line extrapolation of each module's daily figure (Section 3.2);

Sl. No	Module	Avg. Txns/Day	Avg. Object Size	Size/Day	Size/Month	Size/6 Months	Size/12 Months
1	Paper Based Registration (All Districts)	13,000	15.38 MB / txn	200 GB	6 TB	36 TB	72 TB
2	Paperless Registration (1 District)	150	20.00 MB / txn	3 GB	90 GB	540 GB	1.08 TB
3	EC (Encumbrance Certificate)	30,000	100 KB / txn	3 GB	90 GB	540 GB	1.08 TB
4	CC (Certified Copy)	7,000	143 KB / txn	1 GB	30 GB	180 GB	360 GB
5	Marriage Registration	2,000	7.50 MB / txn	15 GB	450 GB	2.70 TB	5.40 TB
6	Firm Registration	100	5.00 MB / txn	0.5 GB	15 GB	90 GB	180 GB

Sl. No	Module	Avg. Txns/Day	Avg. Object Size	Size/Day	Size/Month	Size/6 Months	Size/12 Months
8	e-Stamp	600	333 KB / txn	0.2 GB	6 GB	36 GB	72 GB
	Total — Part I	52,850	15 MB	222.7 GB	6.68 TB	40.09 TB	80.17 TB

5.2 Part II —e- Stamp Roll-out (Yet to Go-Live)

Separation of e-Stamp into an independent module with a projected volume of about 1,00,000 transactions/day, materially higher than the transaction count currently embedded in the live e-Stamp line in Part I (600/day) — reflecting the broader transaction base once e-Stamp issuance is handled independently of registration.

Sl.No	Module	Avg. Txns/Day	Avg. Object Size	Size/Day	Size/Month	Size/6 Months	Size/12 Months
1	e-Stamp (Separation) – Yet to go live	1,00,000	500 KB / txn	50 GB	1.50 TB	9 TB	18 TB

5.3 Part III — Paperless Registration (Yet to Go-Live)

Statewide roll-out across all 30 districts. One district Chamarajanagar is live already.

Sl. No	Module	Avg. Txns/Day	Avg. Object Size	Size/Day	Size/Month	Size/6 Months	Size/12 Months
1	Paperless Registration – All 30 Districts (Yet to go live)	13,000	24.00 MB / txn	312 GB	9.36 TB	56.16 TB	112.32 TB

6. Basis of Estimation — Average Object Size per Transaction

The following derived object sizes support the reasonableness of the daily figures supplied. As a cross-check: the Paperless Registration pilot (1 district) shows an average object size of about 20 MB/transaction, while the statewide Paperless projection (Part II) shows about 24 MB/transaction — the same order of magnitude, as expected for the same document type at larger scale, which supports the consistency of the Part II projection.

Module	Source Part	Avg. Object Size / Transaction
Paper Based Registration	Part I	15.38 MB / txn
Paperless Registration (1 District pilot)	Part I	20.00 MB / txn
Paperless Registration (35 Districts, statewide)	Part II	24.00 MB / txn
EC (Encumbrance Certificate)	Part I	100 KB / txn
CC (Certified Copy)	Part I	143 KB / txn
Marriage Registration	Part I	7.50 MB / txn
Firm Registration	Part I	5.00 MB / txn
NESL integration stream for e-Stamp	Part I	1.00 MB / txn
e-Stamp (current, embedded)	Part I	333 KB / txn

Module	Source Part	Avg. Object Size / Transaction
e-Stamp (post-separation)	Part III	500 KB / txn

7. Consolidated / Phased Requirement

Storage can be provisioned in phases aligned to module go-live or provision upfront, rather a just-in-time allocation. The table below rolls up the net requirement at each phase (Part I, Sl. No. 2 is retired at Phase 2 per the assumption in Section 4).

Phase	Trigger	Net Monthly Requirement	Annualized (12 mo.)
Phase 1 — Current	All Part I modules live, as currently deployed	6.71 TB	80.53 TB
Phase 2	e-Stamp separation goes live (Part III)	17.48 TB	209.77 TB
Phase 3	Paperless Registration goes live statewide (30 districts); single-district pilot	15.98 TB	191.77 TB

At full roll-out (Phase 3), the projected net storage run-rate is approximately 17.48 TB per month, or approximately 209.77 TB on an annualized basis, before applying the overhead and headroom considerations in Section 8 below.

8. Storage Overhead & Headroom Recommendations

The figures in Sections 5–7 represent usable, application-level object storage. The following general considerations should be applied by SDC when translating this into raw capacity to provision, and should be confirmed against the specific Scality cluster configuration and data-protection scheme in use:

Consideration	Typical Range	Applied To
Erasure-coding / data-protection overhead	≈ 1.4x – 1.6x raw-to-usable (scheme-dependent)	Applied by SDC when converting the usable capacity below into raw disk capacity to provision
Operational headroom buffer	7% – 15% of projected usable requirement	Recommended addition to the Phase 3 figure to absorb month-to-month variance
Transaction growth (organic, non-rollout)	Not separately modelled here	To be reviewed periodically against actual transaction trends and re-forecast

These are indicative industry ranges provided to aid provisioning discussions and are not a substitute for sizing confirmation from the SDC/Scality platform team based on the actual cluster's erasure-coding configuration.

9. Risks & Considerations

- Volumes used are averages, not peak-day figures; burst capacity (e.g. campaign-driven registration spikes) is not separately provisioned for.
- Go-live timelines for Part II and Part III are dependent on the respective module roll-out schedules; provisioning should be phased accordingly rather than committed as a single Day-1 allocation.
- Organic transaction growth within already-live modules (independent of the Part II/III roll-outs) is not modelled and should be reviewed periodically against actuals.
- Object size assumptions (Section 6) are derived from current averages and may shift if document quality/resolution standards change (e.g. higher-resolution scanning).

10. Recommendation

It is recommended that:

- Storage be provisioned immediately for Phase 1 and Phase 2 (current live requirement) to eliminate near-term capacity risk.
- Capacity for Phase 3 can be reserved/scheduled in SDC's provisioning plan against the respective module go-live dates.
- The headroom and overhead guidance in Section 8 be applied when converting the usable-capacity figures in this document into a raw-capacity provisioning order.
- This sizing be revisited at each phase gate (and at minimum every 6 months) against actual observed consumption, and re-forecast if actuals diverge materially from the projections in Section 5.